



2.9

Logarithmic Expressions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: $\log_b x$ is the exponent to which b must be raised to obtain x .

2. Write one precise sentence explaining this principle: Logarithm inputs must be positive.

3. Write one precise sentence explaining this principle: Base must be positive and not 1.

4. Write one precise sentence explaining this principle: Exact evaluation is often easiest by rewriting as an exponential equation.

Core

5. Evaluate $\log_3(1/27)$.

6. Solve $\log_5 x = 3/2$.

7. Explain why $\log_2(-8)$ is not real.

8. State what the formula $\log_b x = y \iff b^y = x$ tells you and name any required restriction.

9. State what the formula $\log_b 1 = 0$ tells you and name any required restriction.

10. State what the formula $\log_b b = 1$ tells you and name any required restriction.

Medium

11. Evaluate exactly: $\log_2 32$.

12. Evaluate exactly: $\log_3(1/27)$.

13. Evaluate exactly: $\log_{10} 0.001$.

14. Evaluate exactly: $\ln e^7$.

15. Evaluate exactly: $\log_4 8$.

Hard

16. Solve $\log_5 x = -2$.

17. Find b if $\log_b 81 = 4$.

18. Compare $\log_2 7$ and $\log_3 7$ without decimals.

19. Explain why $\log_b x$ is negative for $0 < x < 1$ when $b > 1$.

Difficult

20. Find all real x such that $\log_x 16 = 4$.

21. Solve $\log_4(x^2) = 3$ and explain domain.

22. Order $\log_2 3$, $\log_3 4$, $\log_4 5$ approximately or by reasoning.

Challenge / HOT

23. Prove $\log_b x = 1/\log_x b$ for valid bases.

24. Find all $x > 0, x \neq 1$ satisfying $\log_x 64 = \log_2 x$.

AP-Style Multiple Choice

25. $\log_b x = y$ means
(A) $x^b = y$
(B) $b^y = x$
(C) $y^b = x$
(D) $b^x = y$
26. Logarithm input must be
(A) negative
(B) positive
(C) integer

(D) less than 1

27. $\log_b 1 =$

(A) -1

(B) 0

(C) 1

(D) b

28. $\log_b b =$

(A) 0

(B) 1

(C) b

(D) undefined

29. $\log_2(1/8) =$

(A) -3

(B) -1/3

(C) 3

(D) 8

30. Base 1 is invalid because

(A) outputs negative

(B) 1^x never produces values other than 1

(C) it is not real

(D) logs require integer base

AP-Style Free Response

31. Without a calculator evaluate: (a) $\log_5 125$, (b) $\log_{1/2} 8$, (c) $\log_9 3$, (d) $\ln(1/e^4)$. Explain each via exponential form.

32. A pH-like scale is $S = -\log_{10} C$. (a) Find S for $C=10^{-6}$. (b) If concentration is multiplied by 100, how does S change? (c) Solve C in terms of S.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.